

NovaStack Technologies Knowledge Base

1. Company Overview

About NovaStack Technologies

NovaStack Technologies is a software company that helps businesses design, build, migrate, and improve digital products.

We work with startups, small businesses, and growing companies that need reliable websites, custom applications, cloud infrastructure, or technical guidance.

Our main services are:

- Web design and development
- Custom application development
- AWS cloud migration
- Technical consultancy
- Support and maintenance

NovaStack focuses on practical digital solutions that are easy to use, reliable, secure, and built around real business needs.

2. Web Design and Development

Website Design Services

NovaStack designs and builds modern websites for businesses and organizations that need a strong online presence.

We create websites that are responsive, easy to navigate, and aligned with the client's brand and goals.

Common website projects include:

- Business websites
- Landing pages
- Portfolio websites
- Event websites
- E-commerce websites
- Blogs and content websites
- SaaS marketing websites

Every website is designed to work well on desktop, tablet, and mobile devices.

Website Redesign

NovaStack also helps clients improve existing websites.

A redesign may be recommended when a website:

- Looks outdated
- Loads slowly
- Is difficult to use on mobile
- Has unclear navigation
- Does not explain the business clearly
- Does not convert visitors into leads or customers

Before redesigning a website, we review the current site, identify weak areas, and recommend improvements based on the client's goals.

UI and UX Design

NovaStack provides UI and UX design for websites, web applications, dashboards, and mobile apps.

This may include:

- Wireframes
- User flows
- Page layouts
- Interface mockups
- Design systems
- Clickable prototypes

Good design helps users understand the product quickly and complete tasks with less friction.

3. Application Development

Custom Application Development

NovaStack builds custom software applications for businesses that need more than a basic website.

These applications are designed around specific business processes, users, and workflows.

Examples include:

- Admin dashboards
- Customer portals
- Booking systems
- Inventory systems
- Reporting systems
- Internal business tools
- SaaS platforms
- Workflow automation systems

Custom applications help businesses reduce manual work, improve visibility, and manage operations more efficiently.

Dashboard Development

NovaStack builds dashboards for teams that need to manage data, users, reports, and business activity from one place.

A dashboard may include:

- User authentication
- Role-based access control
- Tables and filters
- Charts and analytics
- File uploads
- Notifications
- Audit logs
- Exportable reports
- Third-party integrations

Dashboards are useful for internal teams, administrators, operations teams, and customer support teams.

Mobile App Development

NovaStack develops mobile applications for Android and iOS.

Mobile apps may be built for customers, employees, field teams, or service users.

Common mobile app features include:

- User accounts
- Push notifications
- Forms
- Payments
- Chat or messaging
- Location-based features
- Offline access
- API integrations

The development approach depends on the project goals, budget, timeline, and required platform support.

API Development

NovaStack designs and builds APIs that connect websites, mobile apps, dashboards, databases, and third-party services.

API development may include:

- REST APIs
- Authentication
- Authorization
- Database integration

- Payment integrations
- Messaging integrations
- API documentation
- Testing and deployment

Well-designed APIs make applications easier to maintain, extend, and integrate with other systems.

4. AWS Cloud Migration

AWS Cloud Migration Services

NovaStack helps businesses move websites, applications, databases, and infrastructure to Amazon Web Services.

AWS cloud migration can improve reliability, scalability, security, and infrastructure control.

Migration may include:

- Moving websites from shared hosting or VPS servers
- Migrating databases to AWS
- Moving files and media to cloud storage
- Setting up application servers
- Configuring DNS and domains
- Setting up SSL certificates
- Creating backup plans
- Improving deployment workflows
- Adding monitoring and logging

AWS Services We Work With

NovaStack commonly works with AWS services such as:

- Amazon EC2 for virtual servers
- Amazon S3 for file and asset storage
- AWS Lambda for serverless functions
- Amazon RDS for relational databases
- Amazon DynamoDB for NoSQL databases
- Amazon CloudFront for content delivery
- Amazon Route 53 for DNS management
- Amazon API Gateway for APIs
- AWS IAM for access control
- Amazon CloudWatch for logs and monitoring
- AWS Amplify for web app hosting and deployment

The final AWS setup depends on the client's application, traffic, budget, and long-term growth plans.

Website Migration to AWS

NovaStack can migrate existing websites to AWS from shared hosting, VPS providers, or other cloud platforms.

A typical website migration includes:

- Reviewing the current hosting setup
- Preparing the AWS environment
- Moving website files
- Migrating the database
- Configuring SSL
- Updating DNS records
- Testing the website
- Switching live traffic
- Monitoring after launch

The goal is to complete the migration with minimal disruption to users.

Database Migration

NovaStack helps clients migrate databases to AWS services such as Amazon RDS, Aurora, or DynamoDB.

Before migration, we review the current database structure, size, performance needs, and application requirements.

Database migration may require careful planning to avoid data loss and reduce downtime.

Downtime During Migration

NovaStack aims to minimize downtime during cloud migration.

For most migrations, the new AWS environment is prepared and tested before live traffic is switched.

Some projects may require a short maintenance window, especially when moving active databases or production systems.

Clients are informed before any planned downtime.

5. Technical Consultancy

Consultancy Services

NovaStack provides technical consultancy for businesses that need guidance before, during, or after a software project.

Consultancy is useful when a business needs expert advice but may not yet be ready for full development work.

Our consultancy services include:

- Software architecture review
- AWS infrastructure planning
- Cloud cost optimization

- Product planning
- MVP planning
- Security review
- Performance review
- Technology stack recommendations
- Development workflow review
- Vendor and tool selection

Consultancy can be offered as a one-time session, a written review, or ongoing advisory support.

System Review

NovaStack can review an existing software system and provide practical recommendations.

A system review may cover:

- Application architecture
- Code organization
- Database design
- Hosting setup
- Security risks
- Performance bottlenecks
- Deployment process
- Backup strategy
- Logging and monitoring
- Scalability concerns

After the review, NovaStack provides a summary of findings and recommended next steps.

AWS Cost Optimization

NovaStack helps businesses understand and reduce AWS costs.

A cost review may look at:

- Unused resources
- Oversized servers
- Storage costs
- Data transfer costs
- Database usage
- CloudWatch costs
- Lambda usage
- Reserved instance or savings plan opportunities
- Poorly configured infrastructure

The goal is to reduce unnecessary spending without hurting system performance or reliability.

MVP Planning for Startups

NovaStack helps startups plan minimum viable products.

This includes identifying the most important features needed for the first version of the product.

MVP planning may include:

- Feature prioritization
- User flow planning
- Technical architecture
- Development roadmap
- Budget estimation
- Launch planning

The goal is to help startups build a focused first version instead of trying to build everything at once.

6. Project Process

How Projects Work

NovaStack follows a structured project process to help clients move from idea to launch.

The typical process includes:

1. Discovery
2. Planning
3. Design
4. Development
5. Testing
6. Launch
7. Support

Discovery

During discovery, NovaStack learns about the client's business, goals, users, and current challenges.

This stage helps clarify what needs to be built and why it matters.

Planning

During planning, NovaStack defines the project scope, features, timeline, and technical approach.

This stage may include architecture decisions, milestones, and cost estimates.

Design

During design, NovaStack creates the visual and user experience direction for the project.

This may include wireframes, mockups, prototypes, and user flows.

Development

During development, NovaStack builds the website, application, API, or cloud infrastructure.

Clients may receive progress updates throughout the development phase.

Testing

Before launch, NovaStack tests the project to confirm that it works correctly.

Testing may include:

- Functional testing
- Mobile responsiveness testing
- Browser testing
- Performance testing
- Security checks
- User acceptance testing

Launch

During launch, NovaStack deploys the project to the live environment.

Launch may include domain setup, SSL configuration, hosting setup, database setup, and monitoring.

Support

After launch, NovaStack can provide support, maintenance, monitoring, and improvements depending on the client's agreement.

7. Pricing and Quotes

How Pricing Works

NovaStack provides custom pricing because every project is different.

The cost of a project depends on:

- Project scope
- Number of pages or features
- Design complexity
- Required integrations
- Cloud infrastructure needs
- Timeline
- Support requirements
- Existing system condition

A simple website usually costs less than a custom application with dashboards, authentication, payments, reports, and integrations.

Fixed-Price Projects

NovaStack can offer fixed-price projects when the scope is clearly defined.

Fixed pricing works best when requirements are stable and unlikely to change during development.

Monthly Retainers

NovaStack offers monthly retainers for clients who need ongoing support, maintenance, cloud monitoring, improvements, or technical advice.

A monthly retainer may include:

- Bug fixes
- Website updates
- Small feature improvements
- Security updates
- AWS monitoring
- Cloud cost reviews
- Technical consultation
- Performance checks

The retainer scope depends on the client's needs.

Deposits and Milestones

Most projects require an initial deposit before work begins.

The remaining payment may be divided across project milestones, depending on the project size and agreement.

8. Support and Maintenance

Post-Launch Support

NovaStack provides support after websites, applications, and cloud systems go live.

Support may include:

- Bug fixes
- Content updates
- Security updates
- Performance monitoring
- Infrastructure checks
- Backup checks
- Feature improvements
- AWS monitoring

Support terms depend on the maintenance plan or project agreement.

Website or Application Downtime

If a client's website or application is down, they should contact NovaStack support with clear details.

Helpful information includes:

- Website or application URL
- Screenshot of the error
- Time the issue started
- Whether all users are affected
- Any recent changes made
- Error messages, if available

Critical production issues are prioritized for active support clients.

Emergency Support

Emergency support may be available for active clients on selected support plans.

Emergency issues may include:

- Website downtime
- Application outage
- Payment failure
- Security incident
- Cloud infrastructure failure
- Broken customer-facing functionality

Emergency support availability depends on the service agreement.

9. Security

Security Practices

NovaStack follows secure development and deployment practices.

Security measures may include:

- HTTPS setup
- Secure authentication
- Role-based access control
- Input validation
- Environment variable protection
- Database access control
- AWS IAM best practices
- Logging and monitoring
- Backup configuration
- Secure API design

Security requirements vary depending on the type of project and the sensitivity of the data involved.

Password Handling

NovaStack does not store passwords in plain text.

Applications that require passwords should use secure password hashing or trusted authentication providers.

Access Control

NovaStack can implement role-based access control for applications that have different user types.

For example, an application may have:

- Admin users
- Staff users
- Customer users
- Read-only users
- Support users

Access control helps ensure that users only see and perform actions they are allowed to access.

Two-Factor Authentication

NovaStack can add two-factor authentication where required.

Two-factor authentication adds an extra layer of security by requiring users to verify their identity using an additional method.

10. Hosting and Deployment

Hosting Support

NovaStack helps clients set up hosting for websites, applications, APIs, and cloud systems.

We commonly work with:

- AWS
- Vercel
- Netlify
- DigitalOcean
- Cloudflare
- Render

For larger or more complex systems, NovaStack usually recommends AWS.

Deployment Services

NovaStack can deploy websites, APIs, dashboards, and mobile app backends.

Deployment may include:

- Domain setup
- SSL certificate setup
- Database setup
- Environment configuration
- CI/CD pipeline setup
- Storage configuration
- Monitoring setup
- Backup configuration

A good deployment setup helps make future updates safer and easier.

11. Integrations

Payment Integrations

NovaStack can integrate payment systems into websites and applications.

Possible payment providers include:

- Stripe
- PayPal
- Flutterwave
- Pesapal
- Mobile money providers
- Bank payment providers

The best payment provider depends on the client's country, currency, business model, and required payment flow.

Third-Party API Integrations

NovaStack integrates applications with third-party services.

Common integrations include:

- Payment providers
- CRM systems
- Email platforms
- SMS providers
- WhatsApp or messaging platforms
- Analytics tools
- Cloud services
- Accounting tools
- Internal company APIs

Before integration, NovaStack reviews the API documentation, authentication method, limits, and expected behavior.

12. Client Requirements

Information Needed Before Starting a Website Project

Before starting a website project, it helps to prepare:

- Company name
- Logo
- Brand colors
- Website content
- Images
- List of required pages
- Domain access
- Hosting access, if available
- Examples of websites the client likes

If the client does not have everything ready, NovaStack can help plan and organize the project.

Information Needed Before Starting an App Project

Before starting an application project, it helps to prepare:

- Description of the business problem
- Target users
- Main features
- User roles
- Existing systems involved
- Required integrations
- Data to be stored
- Reporting needs
- Budget range
- Expected launch timeline

Clear requirements help reduce delays and avoid unnecessary changes during development.

Information Needed Before Cloud Migration

Before starting AWS migration, NovaStack may request:

- Current hosting details
- Domain access
- Database access
- Application source code
- Server access
- Traffic expectations
- Backup information
- Existing deployment process
- Current infrastructure costs
- Known performance or reliability issues

This information helps NovaStack plan a safer migration.

13. Frequently Asked Questions

Can NovaStack build both the design and the application?

Yes. NovaStack can handle both UI design and application development.

This is useful for clients who want one team to manage the full product experience from design to deployment.

Can NovaStack work with an existing team?

Yes. NovaStack can work with internal teams, freelance developers, designers, or IT departments.

We can support specific parts of a project or provide technical guidance.

Does NovaStack provide source code?

Yes. For custom software projects, clients receive the source code according to the project agreement and payment terms.

Can clients request changes during development?

Yes. Clients can request changes during development.

If a change affects the project scope, cost, or timeline, NovaStack will discuss it before proceeding.

Does NovaStack sign NDAs?

Yes. NovaStack can sign a non-disclosure agreement when required.

Can NovaStack maintain a system built by another developer?

Yes. NovaStack can review and maintain existing systems built by another developer.

Before taking over maintenance, we usually review the codebase, hosting setup, database, and deployment process.

Can NovaStack help choose the right technology stack?

Yes. NovaStack provides technology recommendations based on the project goals, budget, team skill level, performance needs, and long-term maintenance requirements.

Does NovaStack offer training?

Yes. NovaStack can provide training for clients or internal teams.

Training may cover:

- How to use a dashboard

- How to manage website content
 - How to understand AWS basics
 - How to follow a deployment process
 - How to manage users and roles
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14. Contact and Escalation

General Support

Clients can contact NovaStack support for help with websites, applications, cloud infrastructure, bugs, and maintenance requests.

Support email: support@novastack.example.com

Sales and Project Inquiries

New clients can contact NovaStack for project discussions, quotes, and consultancy inquiries.

Sales email: sales@novastack.example.com

Escalation

Urgent production issues for active support clients are escalated based on severity.

High-priority issues include:

- Production downtime
- Security incidents
- Payment failures
- Major user access issues
- Critical business workflow failures

Escalation terms depend on the client's support agreement.